COMPUTATIONAL GEOMETRY · CASE STUDY

Evaluating Utility-Service Accessibility in Dhaka City with Voronoi Diagrams

Partitioning the city by nearest facility to measure how evenly six essential services reach residents — and to locate, quantitatively, where a new facility would help most.

Methodological case study · analysis area 217.3 km² · 75 facilities across 6 services · figures and statistics generated by the accompanying interactive tool

Abstract. A Voronoi diagram assigns every point in a plane to its nearest site. Applied to a city, it turns a scatter of facilities into **service regions** whose size and the distance-to-nearest within them measure accessibility. We compile approximate coordinates for six utility services in Dhaka — metro (MRT-6) stations, hospitals, water (WASA) points, fuel/CNG stations, bus stops and fire stations — build their Voronoi partitions over the city boundary, and score coverage by distance-to-nearest facility. Across all six services, **66–75% of the city lies more than 1.5 km from the nearest facility**, and the consistently underserved belt is the **eastern fringe (Bashundhara–Badda)** and the **northern and north-western periphery (Dakshinkhan, Uttarkhan, Turag)**. Using a largest-empty-circle rule we propose three priority sites per service; several coincide, marking cross-cutting priority zones for investment.

CONTENTS

1 · Introduction & objective

Dhaka is one of the world's densest cities, and its utility services have grown unevenly. Residents in some neighbourhoods reach a hospital, a metro station or a fire station within a few hundred metres; others travel several kilometres. Measuring that inequality objectively — not by anecdote — is a prerequisite for planning where to invest next.

Objective. Evaluate the spatial accessibility of utility services in Dhaka City using Voronoi diagrams, identify well-served and underserved areas, and recommend where new facilities should be added or existing services strengthened. Dhaka is used as the demonstrating case study.

The Voronoi diagram is the natural instrument for this question: it is exactly the geometry of “which facility is closest,” so its cells *are* the service catchments, and the distance to the cell's site *is* the travel burden a resident bears.

2 · Study area & data

The analysis area is a polygon approximating Dhaka's built-up extent (Uttara and Turag in the north to Shyampur and Jatrabari in the south; the Buriganga/Balu belts as rough east–west limits), covering **217.3 km²**. We compiled approximate coordinates for 75 facilities across six services:

SERVICE	FACILITIES	NOTES
Metro (MRT-6)	16	Line-6 stations, Uttara North → Motijheel
Hospitals	15	Major general & specialist hospitals
Water (WASA)	10	Treatment plants & zonal MODS points
Fuel / CNG	12	Filling & CNG stations
Bus stops	14	Terminals & major stops
Fire stations	8	Fire Service & Civil Defence stations

Coordinates are provided as `data/facilities.csv` (and 24 reference neighbourhoods as `data/neighbourhoods.csv`). They are hand-compiled approximations for a methodological study, not an official register — see §7.

3 · Methodology

The pipeline has four stages, each reproducible in the accompanying tool.

3.1 Voronoi partition

For a set of facility sites *S*, the Voronoi cell of site *s* is every location closer to *s* than to any other site. We compute the partition as a fine **distance raster**: the city is sampled on a 420 × 740 grid (~30 m per pixel); each interior pixel is assigned to its nearest facility (the cell it belongs to) and stores its **distance to that facility**. Cell boundaries are the pixels where the nearest-facility label changes — the Voronoi edges.

3.2 Projection & distance

Longitude/latitude are projected equirectangularly about 23.78°N, with `1° lat ≈ 110.6 km` and `1° lon ≈ 101.9 km`, so straight-line distances are in kilometres. These are Euclidean (as-the-crow-flies) distances, an upper bound on accessibility relative to the real (longer) road-network distance.

3.3 Accessibility metrics

Over all interior pixels we report the **mean**, **median** and **worst-case** distance to the nearest facility, and the **share of the city beyond a threshold** (default 1.5 km — a reasonable walk/short-rickshaw catchment). A large Voronoi cell or a high distance both flag underservice.

3.4 Recommending new sites

New facilities are placed by the **largest-empty-circle** rule: the interior point farthest from every existing facility is the location a new facility would relieve most. We apply it greedily — place the site, add it to *S*, recompute, repeat — to propose the top priority sites for each service.

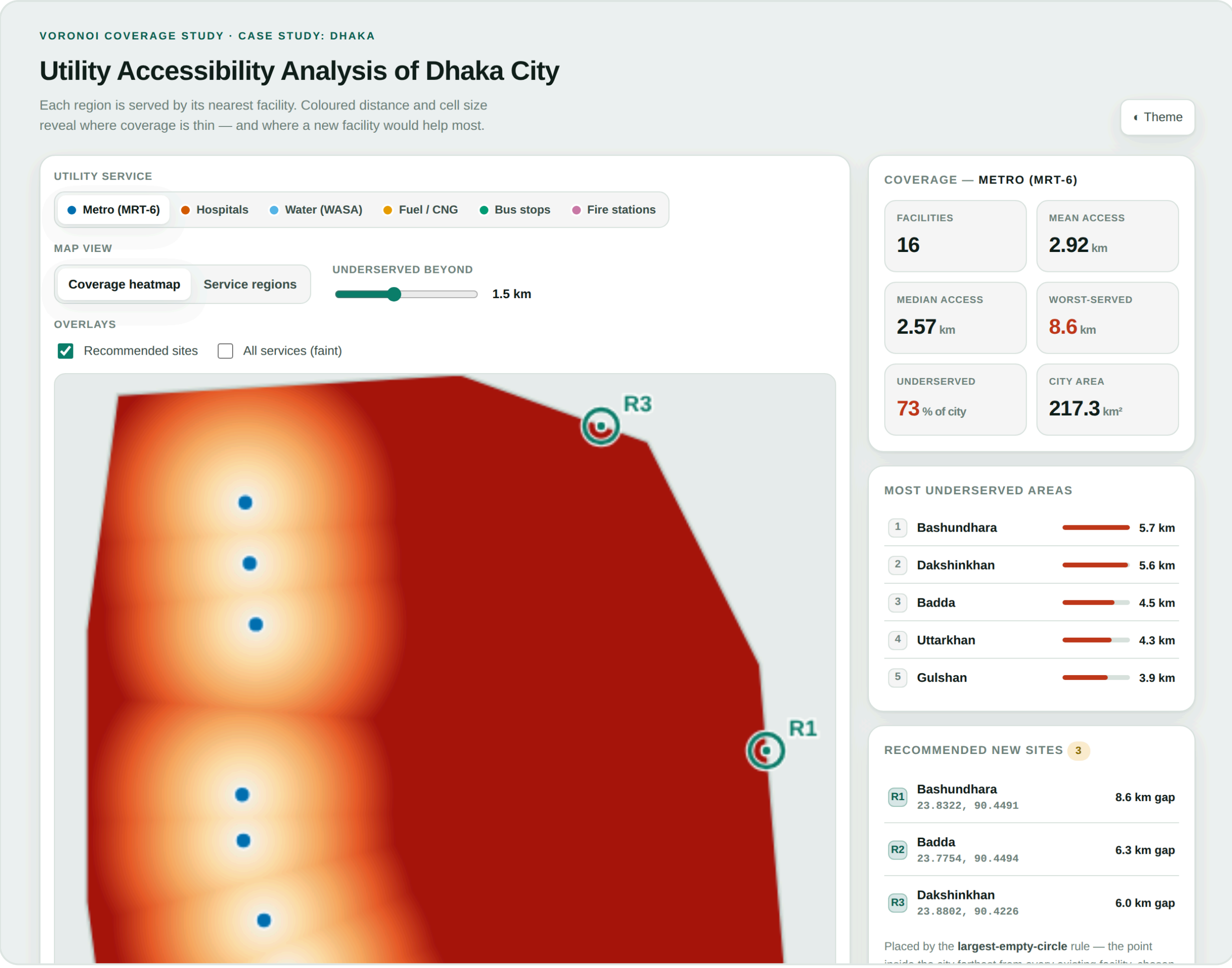


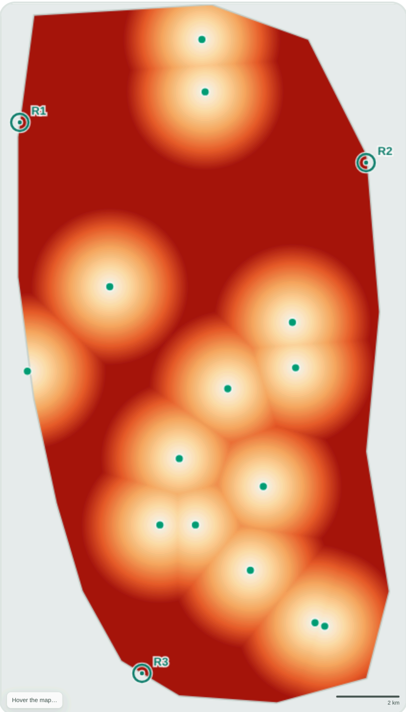
Figure 1. The interactive tool. Left: coverage heatmap for the metro layer, with recommended new sites (R1–R3). Right: live accessibility statistics, the ranked underserved areas and the recommended sites for the selected service. All figures and numbers in this report are produced by this tool.

4 · Results by service

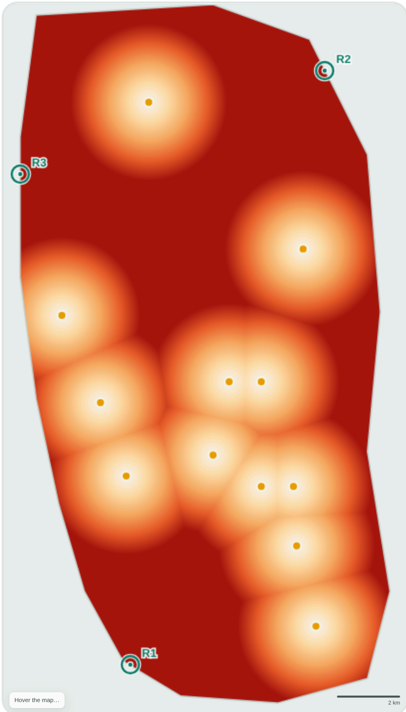
The table ranks services from most to least accessible by mean distance-to-nearest. A striking result: **the metro has the most facilities (16) yet the worst accessibility** — because its stations are *collinear* (a single line), the whole eastern half of the city falls far outside any station's reach. Point-distributed services (bus, fuel) cover area far more evenly with fewer facilities.

SERVICE	FAC.	MEAN KM	MEDIAN KM	90TH PCT	WORST KM	> 1.5 KM
● Bus stops	14	**2.15**	2.01	3.77	5.91	66.0%
● Fuel / CNG	12	2.18	2.05	3.75	5.99	67.6%
● Hospitals	15	2.23	2.09	3.91	6.82	69.7%

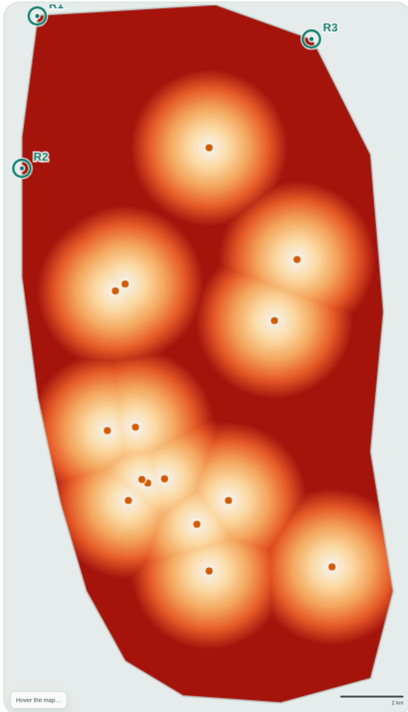
SERVICE	FAC.	MEAN KM	MEDIAN KM	90TH PCT	WORST KM	> 1.5 KM
Water (WASA)	10	2.33	2.15	4.07	7.05	71.7%
Fire stations	8	2.34	2.19	3.99	6.65	74.5%
Metro (MRT-6)	16	**2.92**	2.57	5.87	**8.60**	73.4%



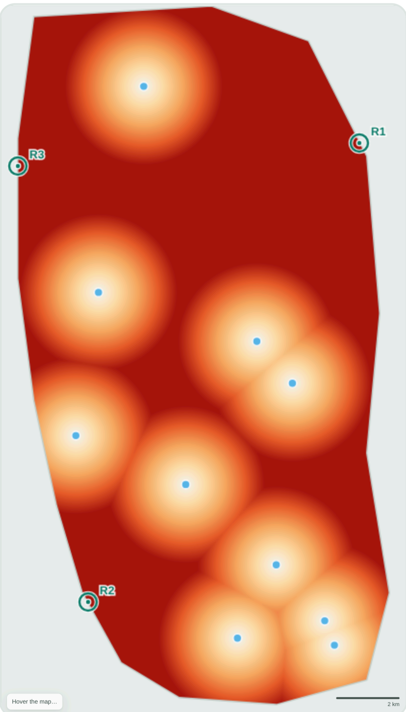
Bus stops



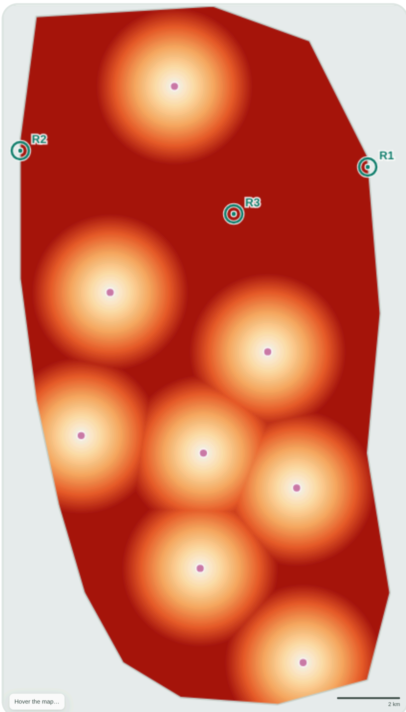
Fuel / CNG



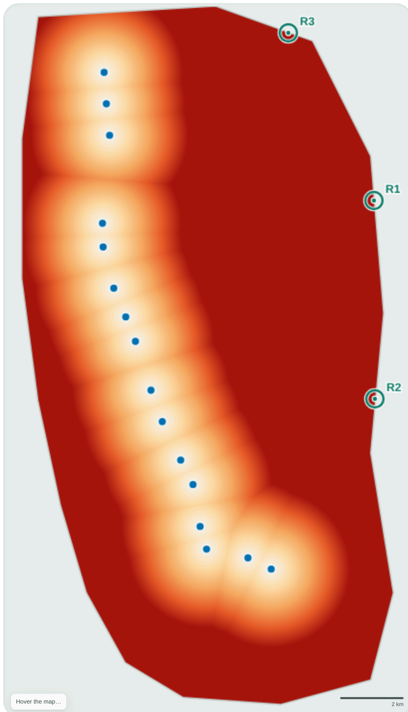
Hospitals



Water (WASA)



Fire stations



Metro (MRT-6)

Figure 2. Coverage heatmaps, all six services. Light = close to a facility (well served); deep red = far (underserved). Teal rings mark recommended new sites. The eastern and peripheral red zones recur across every service.

Worked example — metro regions

Switching the tool to **Service regions** draws the Voronoi cells themselves, shaded by area. The metro's cells fan out enormously toward the east: a handful of stations “own” the entire Gulshan–Badda–Bashundhara side of the city, each resident there kilometres from a platform.

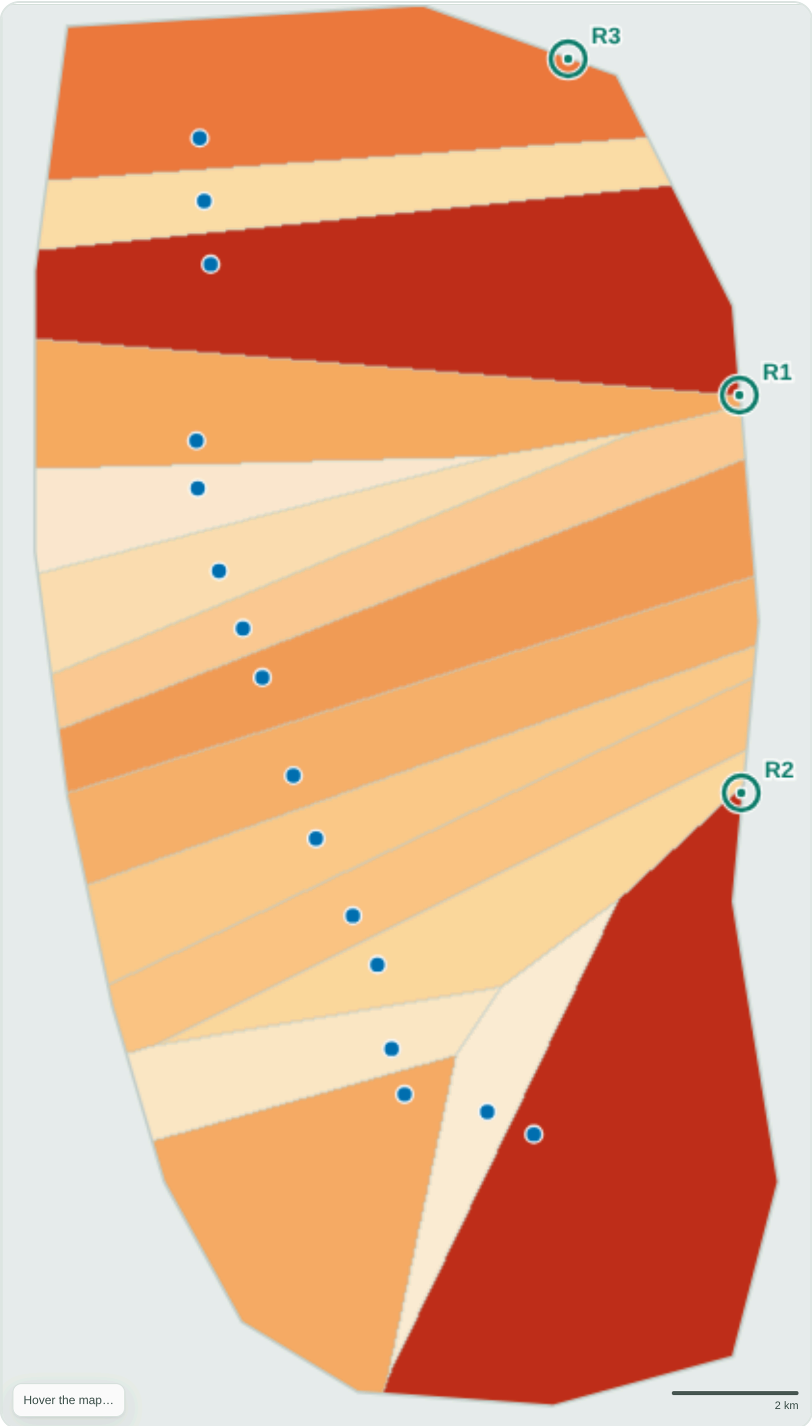


Figure 3. Metro Voronoi regions, shaded by cell area (larger = more underserved). The eastern cells are the largest in the study — the clearest signal of a coverage gap.

5 · Cross-cutting findings

Reading the six maps together, the same places surface as underserved regardless of service:

66–75%	East	Fringe
OF CITY > 1.5 KM FROM NEAREST FACILITY	BASHUNDHARA–BADDA GAP IN 3+ SERVICES	NORTH & NW PERIPHERY PERSISTENTLY FAR

- **Eastern belt — Bashundhara, Vatara, Badda.** The single largest gap. Appears as a top underserved area or recommended site for metro, bus and fire, and is deep-red on every heatmap.
- **Northern fringe — Dakshinkhan, Uttarkhan, Abdullahpur.** Far from metro, hospitals, water and fuel; the built edge has outrun its services.
- **North-west — Turag.** The worst-served point for hospitals, water, bus and fire.
- **South-west — Kamrangirchar, Hazaribagh.** River-hemmed and dense, yet distant from water, fuel, bus and fire.
- **Cantonment & central-north** read as gaps too, though partly because the land is restricted rather than residential.
- **Well served:** the Motijheel–Shahbagh–Dhanmondi–Farmgate core, where every service clusters.

6 · Recommendations

For each service the tool proposes the three highest-impact new sites (largest current gap, in km). The “gap” is the distance from the proposed site to the nearest existing facility — i.e. how far the worst-off resident there travels today.

SERVICE	SITE 1 (GAP)	SITE 2 (GAP)	SITE 3 (GAP)
● Metro	Bashundhara (8.6)	Badda (6.3)	Dakshinkhan (6.0)
● Hospitals	Turag (6.8)	Turag S (4.9)	Dakshinkhan (4.7)
● Water	Dakshinkhan (7.1)	Kamrangirchar (4.8)	Turag (4.7)
● Fuel / CNG	Kamrangirchar (6.0)	Dakshinkhan (5.7)	Turag (4.6)
● Bus	Turag (5.9)	Bashundhara (5.6)	Kamrangirchar (4.7)
● Fire	Bashundhara (6.7)	Turag (5.3)	Cantonment (4.5)

Because the same names recur, we consolidate them into **cross-cutting priority zones** — build once, relieve several services:

Priority 1 **Eastern belt — Bashundhara / Vatara / Badda**

≈ 23.78–23.83 N, 90.42–90.45 E

The largest and most repeated gap. Priority for a metro feeder/extension eastward, plus a fire station and additional bus provision. Fast-urbanising, high-density — the highest-return zone.

Priority 2 **Northern fringe — Dakshinkhan / Uttarkhan / Abdullahpur**

≈ 23.87–23.88 N, 90.40–90.42 E

Recommended for water, fuel, hospital and metro-access improvements — the built edge has grown past its service network.

Priority 3 **North-west & south-west — Turag and Kamrangirchar**

Turag ≈ 23.85 N, 90.34 E · Kamrangirchar ≈ 23.71 N, 90.37 E

Two dense, river-hemmed pockets that are worst-served for several everyday services (hospitals, water, fire, fuel). Small, well-placed additions would sharply cut travel distance.

7 · Limitations

- **Approximate data.** Coordinates are hand-compiled for a methodological demonstration, not surveyed; counts are representative, not exhaustive. Conclusions are about *method and pattern*, not exact metres.
- **Straight-line distance.** Real travel follows roads, rivers and congestion; Euclidean distance under-states true travel time. Rivers and the airport/cantonment also break the “anyone can go anywhere” assumption a plain Voronoi makes.
- **Demand-blind.** The analysis weights area, not population. A large cell over low-density land matters less than a smaller one over a dense settlement; a population-weighted Voronoi is the natural next step.
- **Capacity-blind.** Nearest ≠ adequate — a close but overloaded hospital still under-serves.

8 · The interactive tool

Open `voronoi-dhaka.html` in any browser (double-click — it needs no server or internet). Switch the utility service, toggle between the coverage heatmap and the Voronoi regions, drag the underserved-distance threshold, and hover any point to read its coordinates, nearest facility and distance. Every statistic and figure in this report is generated live by that page from `data/facilities.csv`.

Reproducibility. Method: nearest-facility distance raster over the city polygon (equirectangular projection, 23.78°N) → Voronoi cells + per-pixel accessibility → greedy largest-empty-circle siting. Files: `voronoi-dhaka.html` (tool), `report.html` (this document), `figures/` (maps), `data/facilities.csv`, `data/neighbourhoods.csv`.

Disclaimer. Approximate data for educational analysis. Not an official service record; do not use for operational or emergency planning.